

Seminar 46 - Algebră

Rangul unei matrice. Matrice elementare

Transformări elementare asupra $A \in M_{m,n}(K)$

- permutarea a 2 linii / col ale lui A
- înmulțirea unei linii / col cu $\alpha \in K^*$
- înmulțirea unei linii / col cu $\alpha \in K$ și adunarea la altă

ex: Calculați rang A pentru:

$$i) A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 \\ 1 & 3 & 0 & -3 \end{pmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-r_1}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -2 \\ 0 & 2 & -1 & -4 \end{pmatrix} \xrightarrow{\substack{c_2-c_1 \\ c_3-c_1 \\ c_4-c_1}}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 \\ 0 & 2 & -1 & -4 \end{pmatrix} \xrightarrow{r_3+2r_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -1 & -8 \end{pmatrix} \xrightarrow{r_4-2r_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -1 & -8 \end{pmatrix}$$

$$\Rightarrow \boxed{\text{rang } A = 3}$$

$$ii) A = \begin{pmatrix} 1 & 2 & 1 & -2 \\ 2 & 3 & 1 & 0 \\ 1 & 2 & 2 & -3 \end{pmatrix} \xrightarrow{\substack{r_2-2r_1 \\ r_3-r_1}} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & -1 & 4 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \Rightarrow \text{rang } A = 3$$

$$iii) A = \begin{pmatrix} 3 & 0 & 3 & 0 & 3 \\ 0 & 2 & 0 & 2 & 0 \\ 3 & 2 & 0 & 3 & 2 \\ 0 & 2 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 3 & 0 & 3 & 0 & 3 \\ 0 & 2 & 0 & 2 & 0 \\ 0 & 2 & -3 & 3 & -1 \\ 0 & 2 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{c_4-c_2}$$

$$\sim \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -3 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{rang } A = 3$$

$$10) \quad A = \begin{bmatrix} \boxed{2} & \alpha & -2 & -2 \\ 4 & -1 & 2\alpha & 5 \\ 2 & 10 & 12 & 1 \end{bmatrix}, \alpha \in \mathbb{C}$$

$$\begin{array}{l} l_2 - 2l_1 \\ l_3 - l_1 \end{array} \begin{bmatrix} \alpha & \alpha & -2 & -2 \\ 0 & -1-2\alpha & 2\alpha+4 & 1 \\ 0 & 10-\alpha & -10 & -1 \end{bmatrix} \sim \begin{bmatrix} \alpha & 0 & 0 & 0 \\ 0 & -1-2\alpha & 2\alpha+4 & 1 \\ 0 & 10-\alpha & -10 & -1 \end{bmatrix} \begin{array}{l} c_2 \leftrightarrow c_4 \\ \sim \end{array}$$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 2\alpha+4 & -1-2\alpha \\ 0 & -1 & -10 & 10-\alpha \end{bmatrix} \begin{array}{l} l_3 + l_2 \\ \sim \end{array} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \boxed{2\alpha-6} & 9-3\alpha \end{bmatrix}$$

↓
0

Dass $\alpha = 3 \Rightarrow \text{rang } A = 2$

Dass $\alpha \neq 3 \Rightarrow \text{rang } A = 3$

Def: 0 Matrizenpotenzen resultieren aus Matrizen, welche
 per Operation einer transp. elem. s.m. Matrizen
 elementare.

a) $l_i \leftrightarrow l_j$

$$\begin{array}{c} i \\ j \end{array} \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$\{l_i \leftrightarrow l_j\}$

$$e_i \oplus e_j$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$M_{e_i \oplus e_j} = M_{e_i} \oplus C_j$$

$$\det M_{e_i \oplus e_j} = -1 \Leftrightarrow \exists M_{e_i \oplus e_j}^{-1} = M_{e_i \oplus e_j}$$

$$q \quad e_i \in \alpha e_i, \quad \alpha \in K^*$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \alpha & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$M_{e_i \in \alpha e_i}$$

$$\det M_{e_i \in \alpha e_i} = \alpha \Rightarrow \exists M_{e_i \in \alpha e_i}^{-1} = M_{e_i \in \alpha^{-1} e_i}$$

$$q \quad e_i \leftarrow e_i + \alpha e_j, \quad \alpha \in K$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & \dots & \alpha \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$\Rightarrow \det M_{E_i \leftarrow E_i + \alpha E_j} = 1 \neq 0 \Rightarrow \exists M_{E_i \leftarrow E_i + \alpha E_j}^{-1} = M_{E_i \leftarrow E_i - \alpha E_j}$$

Obs: Inversa unei matrice elementare este o matrice elem

Lemma: Fie $A = (a_{ij}) \in M_{m,n}(K)$. $\nexists$ transf. elem. asupra lui A este rezultat înmulțirii lui A cu o matrice elementară. Mai exact:

• $\nexists$ transf. elem. asupra liniilor lui A se obține prin înmulțirea lui A la stânga cu mat. elem. rezultată prin efectuarea aceleiași transf. asupra lui I_m .

• ——— " ——— coloanelor ——— " ——— la dreapta
 ——— " ——— I_n

Dem:

$$\begin{aligned}
 & \text{a) } \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & & & & & & \\ 0 & 0 & \dots & 0 & \dots & 1 & \dots & 0 \\ \vdots & & & & & & \\ 0 & 0 & \dots & 1 & \dots & 0 & \dots & 0 \\ 0 & 0 & \dots & & & & & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1i} & \dots & a_{1j} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2i} & \dots & a_{2j} & \dots & a_{2m} \\ \vdots & & & & & & & \\ a_{i1} & a_{i2} & \dots & a_{ii} & \dots & a_{ij} & \dots & a_{im} \\ \vdots & & & & & & & \\ a_{j1} & a_{j2} & \dots & a_{ji} & \dots & a_{jj} & \dots & a_{jm} \\ \vdots & & & & & & & \\ a_{m1} & a_{m2} & \dots & a_{mi} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix} \\
 & = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1i} & \dots & a_{1j} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2i} & \dots & a_{2j} & \dots & a_{2m} \\ \vdots & & & & & & & \\ a_{i1} & a_{i2} & \dots & a_{ii} & \dots & a_{ij} & \dots & a_{im} \\ \vdots & & & & & & & \\ a_{j1} & a_{j2} & \dots & a_{ji} & \dots & a_{jj} & \dots & a_{jm} \\ \vdots & & & & & & & \\ a_{m1} & a_{m2} & \dots & & & & & a_{mn} \end{bmatrix}
 \end{aligned}$$

$$b) \begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & \alpha & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & & & \\ a_{i1} & a_{i2} & \dots & a_{im} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix} =$$

$$= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & & & \\ \alpha a_{i1} & \alpha a_{i2} & \dots & \alpha a_{im} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix}$$

$$c) \begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & \dots & 1 & \dots & \alpha & \dots & 0 \\ 0 & 0 & \dots & 0 & \dots & 1 & \dots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & \dots & \dots & \dots & \dots & \dots & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1i} & \dots & a_{1j} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2i} & \dots & a_{2j} & \dots & a_{2m} \\ \vdots & & & & & & & \\ a_{i1} & a_{i2} & \dots & a_{ii} & \dots & a_{ij} & \dots & a_{im} \\ \vdots & & & & & & & \\ a_{j1} & a_{j2} & \dots & a_{ji} & \dots & a_{jj} & \dots & a_{jm} \\ \vdots & & & & & & & \\ a_{m1} & a_{m2} & \dots & a_{mi} & \dots & a_{mj} & \dots & a_{mm} \end{bmatrix} =$$

$$= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1i} & a_{1j} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2i} & a_{2j} & \dots & a_{2m} \\ \vdots & & & & & & \\ a_{i1} + \alpha a_{j1} & \dots & \dots & a_{ii} + \alpha a_{ji} & \dots & \dots & \\ a_{j1} & a_{j2} & \dots & a_{ji} & a_{jj} & \dots & a_{jm} \\ \vdots & & & & & & \\ a_{m1} & a_{m2} & \dots & \dots & \dots & \dots & a_{mm} \end{bmatrix}$$

Corolar: O matrice inversibilă este un produs de matrice elementare

dem: Fie $A \in GL_n(K) \Rightarrow$ printr-o succesiune de transf. elem. obținem I_n

deci $\exists E_1, E_2, \dots, E_q$ matrice elem de ordin n , $q \geq 1$ o.

$$A = E_2 E_{2-1} \dots E_1 A = I_m | A^{-1}$$

$$A^{-1} = E_2 E_{2-1} \dots E_2 E_1 \Rightarrow A = (E_2 E_{2-1} \dots E_2 E_1)^{-1} \\ = E_1^{-1} \cdot E_2^{-1} \dots E_2^{-1} \\ \uparrow \quad \uparrow \quad \quad \uparrow \\ \text{matrix elem}$$

$$A \xrightarrow[\text{elem}]{\text{transf}} I_m$$

$$(A | I_m) \sim \text{---} \sim (I_m | A^{-1})$$

ex: Calculate inverse with matrix (dof 3)

$$a) A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 4 & 1 & 4 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & -1 & 1 & 0 & 1 & 0 \\ 4 & 1 & 4 & 0 & 0 & 1 \end{array} \right) \xrightarrow[l_3 - 4l_1]{l_2 - 2l_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -3 & -2 & 1 & 0 \\ 0 & -3 & -4 & -4 & 0 & 1 \end{array} \right) \xrightarrow[l_3 - l_2]{l_1 + \frac{1}{3}l_2}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & -3 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -2 & -1 & 0 \end{array} \right) \xrightarrow[l_1 + l_3]{l_2 - 3l_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{3} & -\frac{2}{3} & 1 \\ 0 & -3 & 0 & 4 & 4 & -3 \\ 0 & 0 & -1 & -2 & -1 & 0 \end{array} \right) \xrightarrow[-l_3]{-\frac{1}{3}l_2}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -5/3 & -2/3 & 1 \\ 0 & 1 & 0 & -4/3 & -4/3 & 1 \\ 0 & 0 & 1 & 2 & 1 & -1 \end{array} \right) \Rightarrow A^{-1} = \begin{pmatrix} -5/3 & -2/3 & 1 \\ -4/3 & -4/3 & 1 \\ 2 & 1 & -1 \end{pmatrix}$$

$$b) A = \begin{pmatrix} 3 & 4 & 2 \\ 6 & 8 & 5 \\ 9 & 12 & 10 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 3 & 4 & 2 & 1 & 0 & 0 \\ 6 & 8 & 5 & 0 & 1 & 0 \\ 9 & 12 & 10 & 0 & 0 & 1 \end{array} \right) \xrightarrow[l_3 - 3l_1]{l_2 - 2l_1} \left(\begin{array}{ccc|ccc} 3 & 4 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 4 & -3 & 0 & 1 \end{array} \right) \Rightarrow \nexists A^{-1}$$

$$c) A = \begin{pmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 2 \end{pmatrix} (\alpha \in \mathbb{R})$$

$$\left(\begin{array}{ccc|ccc} 2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & \alpha & 0 & 0 & 1 \end{array} \right) \xrightarrow{\frac{1}{2}l_1} \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & \alpha & 0 & 0 & 1 \end{array} \right) \begin{array}{l} l_2 - l_1 \\ l_3 + l_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & -2 & -\frac{3}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 3 & \alpha + \frac{3}{2} & \frac{1}{2} & 0 & 1 \end{array} \right) \xrightarrow{-\frac{1}{2}l_2} \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} & -\frac{1}{2} & 0 \\ 0 & 3 & \alpha + \frac{3}{2} & \frac{1}{2} & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} l_1 - l_2 \\ l_3 - 3l_2 \end{array} \sim \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} & -\frac{1}{2} & 0 \\ 0 & 0 & \alpha - \frac{3}{4} & -\frac{1}{4} & \frac{3}{2} & 1 \end{array} \right)$$

$$\text{Does } \alpha = \frac{3}{4} \Rightarrow \nexists A^{-1}$$

$$\alpha \neq \frac{3}{4} \Rightarrow$$